

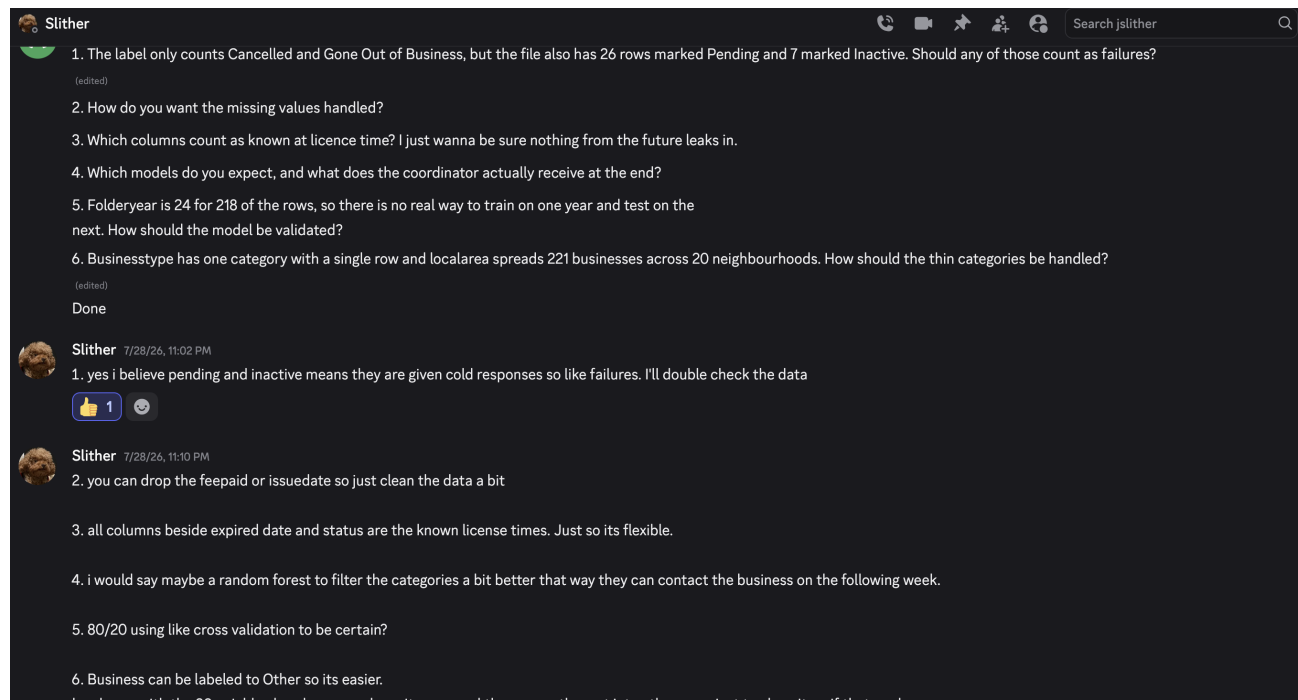
IAT 461 Final Project, Phase 2: Client and Data Scientist Agreement

Jason Salonga and Mahdi Taziki

IAT 461, Summer 2026

July 26, 2026

1. Our conversation



2. The questions I asked as the data scientist and the answers Jason gave me

2.1 The label only counts Cancelled and Gone Out of Business, but the file also has 26 rows marked Pending and 7 marked Inactive. Should any of those count as failures?

Jason said he believes Pending and Inactive both mean the business went cold on the city, so he wants them treated as failures too, and he is going to double check the data to confirm. I flagged that this takes the label from 20 failures out of 221 to 53, so about 24 percent instead of 9. More positive cases is easier to model, but the two versions are not comparable, so I will build against the wider label and report the narrow one as a check.

2.2 How do you want the missing values handled?

He is fine with dropping the columns that have the biggest gaps instead of filling them in, and named feepaid and issueddate. Feepaid is missing for about 53 percent of rows so it goes. Issueddate is only missing for 22 percent, so I will try keeping it with a flag on the blanks and drop it if it does not earn its place. No rows get deleted either way.

2.3 Which columns count as known at licence time? I just want to be sure nothing from the future leaks in.

Everything except expireddate and status counts as known at licence time, and he wants to keep it flexible rather than lock the list down now. Status has to go anyway since the label comes from it. I also flagged that licencenumber and businessname are identifiers, not features, so they stay out of the model and only sit in the output file so rows can be matched back to a business.

2.4 Which models do you expect, and what does the coordinator actually receive at the end?

He asked for a random forest, since it handles the categories better and what he wants out of it is a filtered list the coordinator can work through the following week. The deliverable is a ranked CSV with business name, type, neighbourhood and a risk score, sorted highest risk first, with the top group marked as that week's contact list. I will run a logistic regression first as a baseline to compare against, which he was fine with as long as the forest is what gets handed over.

2.5 Folderyear is 24 for 218 of the rows, so there is no real way to train on one year and test on the next. How should the model be validated?

He suggested an 80 to 20 split with cross validation on top to be more certain, so that is the plan. The split will be stratified so the failures land on both sides, and I will cross validate inside the training portion instead of tuning against the held out part. I did mention that 20 percent of 221 rows is only about 44 businesses, so we agreed I would also report cross validation over the full dataset as a steadier number.

2.6 Businesstype has one category with a single row and localarea spreads 221 businesses across 20 neighbourhoods. How should the thin categories be handled?

For business type he wants the rare ones relabelled as Other so the model is not learning from a single example. For neighbourhood he wants the ones with a decent number of businesses kept as they are and the rest grouped into an other areas bucket. I will set the cut off from the counts and list which ones got merged in the report.

3. The time estimate and workload Jason set as the client

I asked him to put numbers on the workload, since the proposal did not say anything about effort. His estimate is 20 to 25 hours over the three weeks we have left, so about 7 or 8 hours a week. He split it as 5 hours on cleaning the data, 4 on the exploratory figures, 6 on the models, 3 on evaluation, and 5 on the write up and slides.

The checkpoints are the cleaned dataset and figures by July 31, a first working model with validation results by August 5, the random forest and the ranked contact list by August 10, and a draft report and slides by August 14. He also wants a short update twice a week, a reply within a day, and no new questions added after August 5 unless something else comes off the list.

4. What I set as the client on my own project

Jason is the data scientist on my proposal, the Aquila auto-resolve project, so I gave him numbers from the other side. My estimate is 25 to 30 hours of work, and the reason it is higher than his is that my brief has three parts rather than one. If time gets tight I want the battle outcome model done properly first, the strategy comparison second, and the unit role audit trimmed to one clustering run with a short write up.

He asked for some flexibility on timing because of things going on outside the course, so we agreed on the order of the milestones instead of pinning them to dates. That order is the first results, then a model beating the current 63 percent baseline, then a tuned version, then a draft of the report and slides, and we will fix the actual dates once he knows how much room he has. Everything else still holds, meaning regular updates, replies within about a day, and a notebook with a fixed seed and no hard coded paths.

5. Agreement

We both agree that this is what we discussed and that the hours, milestones and check in rules in sections 3 and 4 are what we are each committing to. The one item still open is Jason confirming that Pending and Inactive count as failures, and he will get back to me on that before I finish the cleaning step.

Jason Salonga

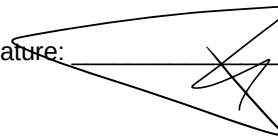
Date: July 26, 2026

Signature: _____

Mahdi Taziki

Date: July 26, 2026

Signature: _____

A handwritten signature in black ink, consisting of a large, stylized 'M' followed by a series of loops and a final cross-like stroke.